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Endogeneity

Statistical Significance, Impact, and Relevance

PhD Course, Prof. Dr. Dr. h.c. Sönke Albers
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Introduction to Endogeneity

What is Endogeneity ?

- Yout want to do valid inference on a target parameter of interest in a general model like:

$$Y = g(X) + \xi$$

- **Endogeneity is Violation of the Exogeneity Assumption** 💡:

$$\mathbb{E}[\xi|X] \neq 0$$

- **Without Exogeneity, we are missing**:
  - $\mathbb{E}[\xi] = 0$ (Law of total expectation)
  - ($\mathbb{E}[X'\xi] = 0$, the regressors are uncorrelated with the errors)

What is Endogeneity ?

- **Endogeneity** refers to situations where an explanatory variable is correlated with the error term in a regression model.
- It is a critical issue because it leads to **biased estimates**, impacting the validity of causal inferences (Ebbes, Papies, and Heerde 2021).

Important

- Sources of ~~evil~~ **endogeneity** (Hill et al. 2021):
  - **1. Omitted variable bias**
  - **2. Measurement error**
  - **3. Simultaneity**.

Why does Endogeneity Matter?

- **Statistical Significance**: Endogeneity can influence the effect size (regression coefficient value).
  - **Causal Inference**: Without addressing endogeneity, any causal claims are questionable.
  - In economics the effect sizes is very important.
- **Marketing Research**: Marketing models often involve simultaneous decisions by consumers and firms, introducing endogeneity (Papies, Ebbes, and Feit 2022).

$\Rightarrow$ "*Everything is endogenous, except for God.*"

Examples of Endogeneity in Marketing

- In many cases one is interested in understanding influences (*causal effects*) of an action (new webdesign, coupons, etc.) on an outcome (sales, KPIs, etc.)
  - Needs valid inference on a parameter of interest
- In marketing, **price** is often endogenous because it is determined simultaneously with demand.
  - e.g. estimating price elasticity of demand
- **Advertising** may also be endogenous due to unobserved factors influencing both sales and ad expenditures (Rossi 2014a).

Overview

Awareness (≈ 1990-2000)

- **IV Methods:**
  - *Moore et al. (1991)* – Early IV methods.
  - *Boulding et al. (1994)*
  - *Villas-Boas & Winer (1999)*

Takeoff (≈ 2000-2010)

- **IV Applications:**
  - *Chintagunta (2001)*, *Sudhir (2001)*
  - *Rust et al. (2002)*
  - *Verhoef (2003)*
- **Methods**:
  - *Petrin & Train (2010)* - Control Function
- **IV Skepticism:**
  - *Bound et al. (1995)* – Weak Instruments

Diversification (≈ 2010-?)

- **Instrument-Free Methods:**
  - *Ebbes et al. (2005)* – Latent IVs
  - *Park & Gupta (2012)* – Gaussian Copulas
- **Quasi-experimental Methods:**
  - *Goldfarb & Tucker (2011)* - Diff-in-Diff
  - *Tirunillai & Tellis (2017)*- Synthetic Controls
  - *Hartmann et al. (2011)* - Regression Discontinuity

1

1. Own illustration based on Papies, Ebbes, and Feit (2022)

Overview

- **Instrumental Variables**: Identify causal effects by using exclusion restrictions, meaning they rely on certain variables that influence the independent variable but not the outcome directly. Examples include methods like two-stage least squares regression and the control function approach, both of which depend on these exclusion restrictions.
- **Gaussian Copula**: This method is used to model the joint relationship between an endogenous variable (one influenced by other factors within the model) and the error term. It provides a way to estimate causal effects when traditional methods may not apply.
- **Latent IVs**: This approach is used when typical instrumental variables are not available. It helps separate the endogenous and exogenous components of a variable by using latent (unobserved) factors.
- **DiD**: This method compares the changes in outcomes between a group that receives a treatment and a control group that does not, using panel data collected before and after the treatment to estimate the causal effect.
- **Synthetic Control**: This approach constructs a synthetic control group as a weighted combination of untreated units, enabling comparisons between treated and untreated groups when an exact match for a control group is unavailable.
- **Regression Discontinuity**: This method compares units that are just above or below a certain threshold used to assign a treatment. It leverages the cutoff point for treatment assignment to estimate the causal effect, often using natural divisions like geographic boundaries or other criteria.

$\Rightarrow$ Good News: We focus on IV-Methods and Copulas for today 😀

A brief insight into the Theory 

Omitted Variable Bias

Our goal is to do valid inference for a target parameter of interest β. Consider the following simple linear regression model (the *long* model) including all covariates:

$$Y = \alpha^{\mathrm{l}} + \beta^{\mathrm{l}} X + \gamma W + \xi, \quad \mathbb{E}[\xi | X] = 0$$

Where:

- Y is the dependent variable (e.g., sales),
- X and W are independent variables (e.g., advertising expenditure, product quality),
- ξ is the error term, capturing all other factors affecting Y.

Omitted Variable Bias

Assume you forgot the other relevant variable W that is correlated both with Y and X and W now is **omitted** from the model. The new (*short* model) specification would be:

$$Y = \alpha^{s} + \beta^{s} X + \zeta, \quad \mathbb{E}[\zeta | X] \neq 0$$

However, when W is omitted from the regression, the error term ζ from the *short* model will include the effect of W:

$$\zeta = \xi + \gamma W$$

This causes ζ to be **correlated** with X, leading to **omitted variable bias**. In this case, the estimate of β will be biased because part of the effect of W on Y will incorrectly be attributed to X.

Omitted Variable Bias

The problem now is the biased estimate of β. In the *short model*, β^{s} is

$$\beta^{\mathrm{s}} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}$$

Substituting Y from the true / *long* form brings us to:

$$\beta^{\mathrm{s}} = \frac{\mathrm{Cov}(X, \alpha^{\mathrm{l}} + \beta^{\mathrm{l}} X + \gamma W + \xi)}{\mathrm{Var}(X)}$$

Omitted Variable Bias

Application of the properties of (co-)variances leads to simplified decomposition:

$$\beta^{\mathrm{s}} = \frac{\overbrace{\mathrm{Cov}(X, \alpha^{\mathrm{l}})}^{=0} + \beta^{\mathrm{l}} \overbrace{\mathrm{Cov}(X, X)}^{=\mathrm{Var}(X)} + \gamma \mathrm{Cov}(X, W) + \overbrace{\mathrm{Cov}(X, \xi)}^{=0}}{\mathrm{Var}(X)}$$

where $\mathrm{Cov}(X, \xi) = 0$ because of $\mathbb{E}[\xi | X] = 0$. We get the simplified form:

$$\beta^{\mathrm{s}} = \beta^{\mathrm{l}} + \gamma \frac{\mathrm{Cov}(X, W)}{\mathrm{Var}(X)}$$

Omitted Variable Bias

We want an unbiased estimate, i.e., for $\beta^{\mathrm{s}} = \beta^{\mathrm{l}}$. Thus, the second term on the right-hand side must vanish:

$$0 = \gamma \frac{\mathrm{Cov}(X, W)}{\mathrm{Var}(X)}.$$

Since $\mathrm{Var}(X) \neq 0$ and $\gamma \neq 0$ by model specification, the product can only be zero if X and W are uncorrelated. Therefore, we require:

$$\mathrm{Cov}(X, W) = 0.$$

Or, in other words, X and W must be orthogonal: $X \perp W$.

Simultaneity

What happens if the variables are linked in a more complex way, so that we need at least two or more equations to describe them?

Example: Price Elasticity of Demand

The price elasticity of demand measures the responsiveness of quantity demanded to changes in price, represented as:

$$\epsilon = \frac{\%\Delta Q}{\%\Delta P} = \frac{\Delta Q}{Q}\frac{P}{\Delta P}$$

- Intersections of demand and supply functions, **simultaneous causality**.

Simultaneity

We can estimate the price elasticity using the least squares approach based on observed prices and quantities. The logarithmic model is:

$$\ln(Q_i) = \alpha + \beta \ln(P_i) + u_i$$

Where:

- Q_i is the quantity,
- P_i is the price,
- u_i is the error term,
- β gives us the elasticity of demand.

Assume unobserved demand and supply shocks:

$\Rightarrow$ Endogeneity of $\ln(P)$

Simultaneity

- Observed equilibrium quantities and prices: Intersections of demand and supply functions
- A simultaneous shift in both the demand and supply functions can lead to **endogeneity** in the observed price $\ln(P_i)$, causing biased estimates if not properly addressed.

$\Rightarrow$ ⚡ Positive slope of the estimated demand curve

▶ Code

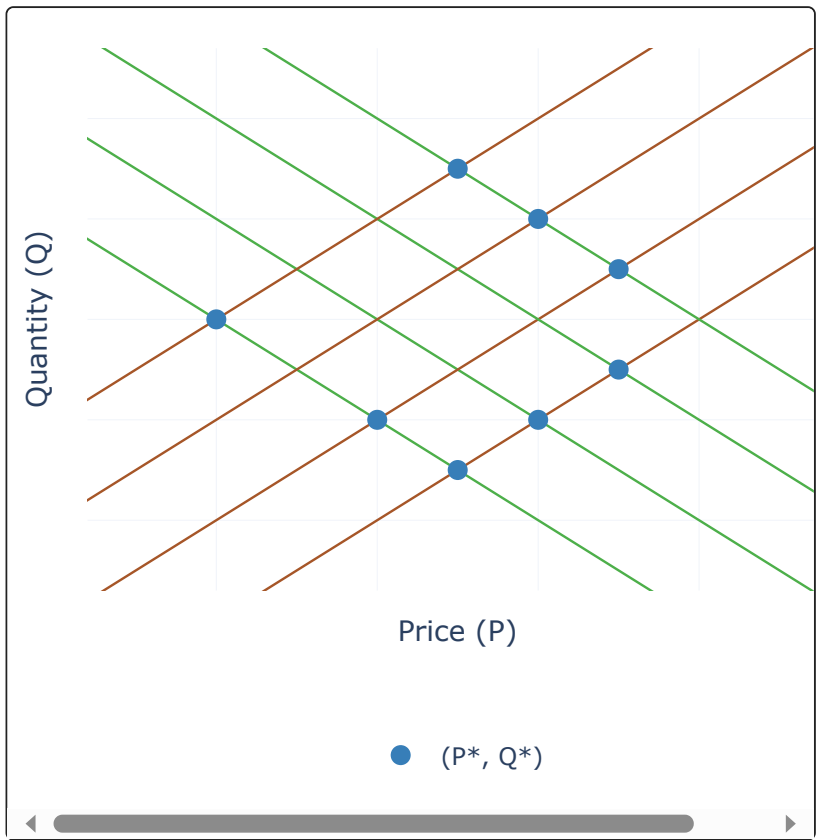


Simultaneity

- Observed equilibrium quantities and prices: Intersections of demand and supply functions
- A simultaneous shift in both the demand and supply functions can lead to **endogeneity** in the observed price $\ln(P_i)$, causing biased estimates if not properly addressed.

$\Rightarrow$ ✅ Negative slope of the demand function

▶ Code



Dealing with Endogeneity 

General

- For the practical approach, we create a simulated data set that is used to briefly illustrate the methods (proof of concept)[1]:

$$
\begin{aligned}
X &\sim \mathcal{N}(0, 1) \\
Z &\sim \mathcal{N}(0, 1) \\
\xi &\sim \mathcal{N}(0, 1)
\end{aligned}
$$

$$
\begin{aligned}
W &= \mathrm{snrW} \times (\beta_Z Z + \beta_\xi \xi) + (1 - \beta_Z - \beta_\xi)\epsilon, \quad \epsilon \sim \mathcal{N}(0, 1) \\
Y &= \alpha + \beta X + \gamma W + \xi
\end{aligned}
$$

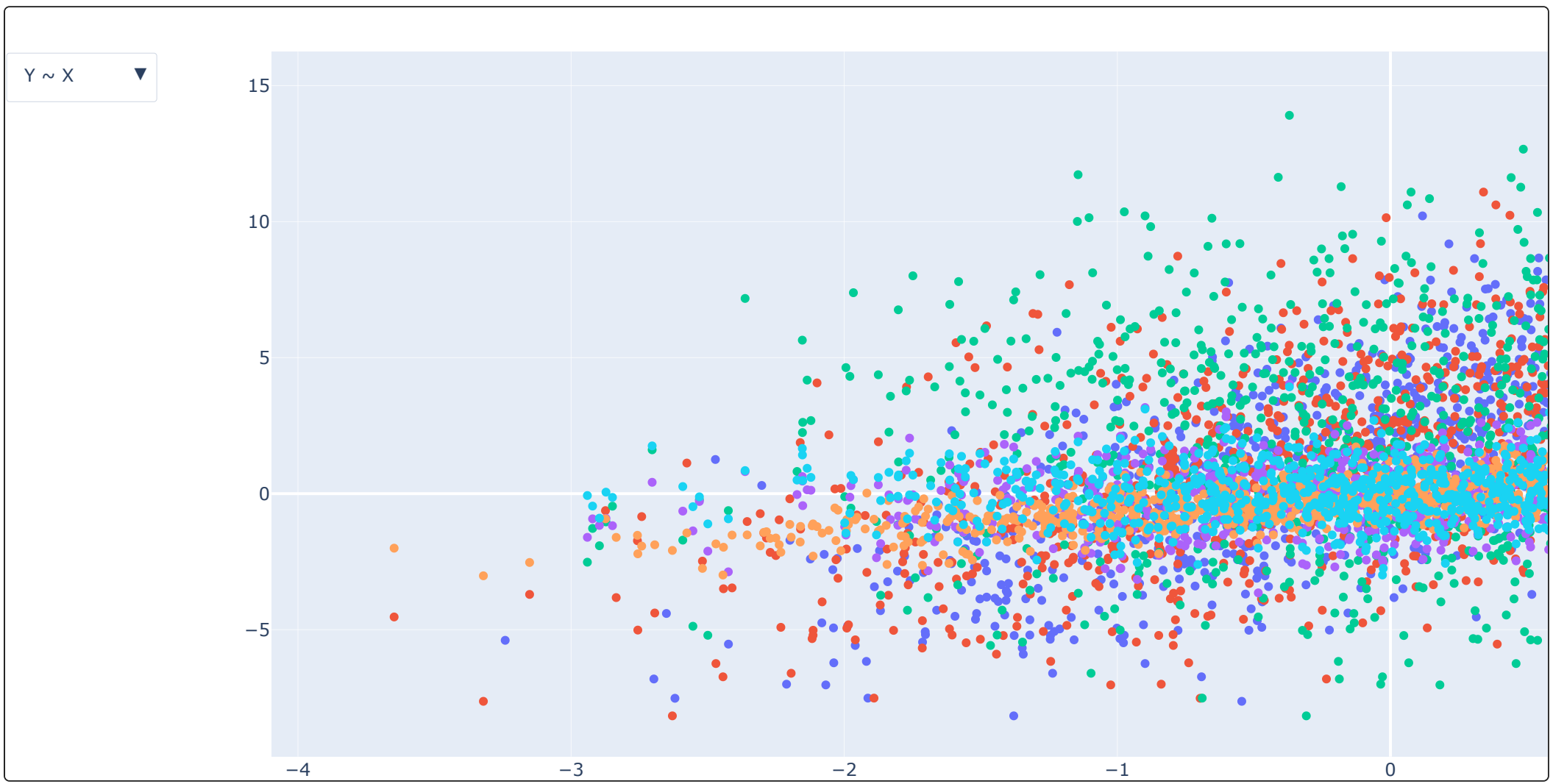
with $\alpha = 2, \beta = 3, \gamma = 1.5, \beta_Z = 0.1, \beta_\xi = 0.3$, and $\mathrm{snrW} = 3$.

▶ Code

1. For each method we look at the practical approach, in this case we use **Python** with the packages `numpy`, `statsmodels` and `linearmodels`.

General

▶ Code



General - OLS Validation

▶ Code

OLS Regression Results			
Dep. Variable:	Y	R-squared:	0.977
Model:	OLS	Adj. R-squared:	0.977
Method:	Least Squares	F-statistic:	2.130e+04
Date:	Mon, 16 Sep 2024	Prob (F-statistic):	0.00
Time:	12:50:10	Log-Likelihood:	-889.10
No. Observations:	1000	AIC:	1784.
Df Residuals:	997	BIC:	1799.
Df Model:	2		
Covariance Type:	nonrobust		

▶ Code

	coef	std err	t	P>|t|	[0.025	0
Intercept	1.9946	0.019	106.953	0.000	1.958	2
X	3.0227	0.019	158.678	0.000	2.985	3
W	2.2041	0.017	132.117	0.000	2.171	2

Instrumental Variables (IVs)

- Instrumental Variable Estimation: General approach for consistent estimation even in the presence of Endogeneity.
  - Even the form of the structural equations is unknown 🥳!
- But beware! We estimate here the **Local** Average Treatment Effects (LATE): Imbens and Angrist (1994) show demonstrate that the IV estimator under weak conditions is a weighted average of LATE.
- Assume a simple linear Example:

$$Y = \alpha + \beta X + \gamma W + \xi$$

Where

- X is exogenous (holding orthogonality condition $\mathbb{E}[X'\xi] = 0$).
- W is an endogenous variable.
- and ξ as the errorterm.

Instrumental Variables (IVs)

💡 Idea: IV Estimation[1]

- Search for instrumental variables, Z, i.e., exogenous variables that are correlated with the endogenous regressor W
  - Conditions for a Valid Instrumental Variable:
    1. **IV1. Relevance**: $\text{Cor}(Z, X) \neq 0$ called the *relevance criterion*.
    2. **IV2. Exogeneity**: $\text{Cor}(Z, \xi) = 0$ called the *exclusion restriction*
- Isolate the part of the variation in W that is not correlated with ξ
- Use the exogenous variation in W to estimate the regression coefficients

$\Rightarrow$ For linear models, we can follow the simple **Two-Stage Least Squares (TSLS)** approach!

1. Ebbes, Papies, and Heerde (2021)

IVs in Marketing 📈

Managers' Behavior is Source of Endogeneity

Possible IV Cases and IVs for Marketing, taken and slightly changed from Ebbes, Papies, and Heerde (2021)

Dependent Variable	Potentially Endogenous Regressor	Potential Causes for Endogeneity	Potential IVs (with citations)
Demand	Price	Managers set prices based on factors that are related to demand but are unobserved by the researcher: - Promotional activities, Changes in popularity, Product characteristics (for cross-sectional analyses)	Costs of ingredients and transportation (Rooderkerk et al. 2013, p. 706); Prices in other markets (Nevo 2001, p. 320)
Demand	Advertising	Managers set advertising based on factors that are related to demand but are unobserved by the researcher: - Price changes, Changes in preferences or popularity, (Expected) sales volume	Advertising expenditures of other (noncompeting) firms (Dinner et al. 2014, p. 534); Advertising cost
Demand	Detailing	Managers set detailing based on factors that are related to demand but are unobserved by the researcher, e.g., large accounts receive more detailing	Costs for detailing (e.g., wages; Chintagunta and Desiraju 2005, p. 74)
Firm Performance	CMO Presence	Firms strategically choose to have a CMO, and this choice depends on unobserved firm characteristics as well as expected performance effects	CMO prevalence among peer firms (Germann et al. 2015, p. 8)

IVs in Marketing 📈

Consumers' Behavior is Source of Endogeneity

Possible IV Cases and IVs for Marketing, taken and slightly changed from Ebbes, Papies, and Heerde (2021)

Dependent Variable	Potentially Endogenous Regressor	Potential Causes for Endogeneity	Potential IVs (with citations)
Demand	Piracy	Unobserved factors (e.g., popularity) affect both a music album's sales as well as piracy	RIAA announcement to start legal actions against file sharing (Bhattacharjee et al. 2007, p. 1364)
Demand (offline channel)	Demand (online channel)	Preferences for channels are correlated	Ease of access to online channel (Gentzkow 2007, p. 726)
Consumer Share of Wallet	Membership in Loyalty Program	Consumers with high intrinsic preference for a firm have a higher share of wallet and are more likely to join the firm's loyalty program	General inclination to join a loyalty program (Leenheer et al. 2007, p. 36)

Instrumental Variables (IVs)

Two-Stage Least Squares (TSLS)

1 First Stage

Decomposition of W into two components:

$$W = \underbrace{\pi_0 + \pi_1 Z}_{\text{Part of } W \text{ that is not correlated with } \xi} + \underbrace{\zeta}_{\text{Part of } W \text{ that is correlated with } \xi}$$

Estimate this equation using OLS: Use the predicted values $\hat{W} = \hat{\pi}_0 + \hat{\pi}_1 Z$ instead of W in the second stage.

- **Relevance**: $\pi_1 \neq 0$
- **Exogeneity**: $\text{Cor}(\pi_0 + \pi_1 Z, \xi) = 0$

Two-Stage Least Squares (TSLS)

2 Second Stage

Estimate a regression model for Y based on the predicted values for $\hat{W}$:

$$Y = \alpha + \beta X + \gamma \hat{W} + \tilde{\xi}$$

- Under the IV assumptions **IV1.** and **IV2.**, the TSLS estimator $\hat{\gamma}^{TSLS}$ is a consistent, unbiased, and asymptotically normally distributed estimator for γ.
- Adjustment of standard errors for $\hat{\gamma}^{TSLS}$ is necessary for valid test statistics and confidence intervals.

Practical Guidelines for Using TSLS[1]

Reporting Results

- When using **2SLS**, researchers should present both the Ordinary Least Squares (OLS) and 2SLS results.
- This includes showing the coefficients and standard errors for each approach.
- Check if the direction of bias in OLS matches the theoretical expectations.
    - In cases where upward bias is expected, such as price effects, TSLS should show a more negative estimate than OLS (Ebbes, Papies, and Heerde 2021).

Over-Identification Test

- When there are more IVs than endogenous regressors, a **Sargan test** (or Hansen J-test) should be performed.
    - This test checks if the IVs are valid (exogenous) by testing their relationship to the residuals from the second-stage regression.
    - Large p-values suggest that the IVs are valid, but this test should not replace theoretical reasoning (Wooldridge 2010; Bascle 2008).

1. according to Ebbes, Papies, and Heerde (2021)

Practical Guidelines for Using TSLS

First-Stage Regression

- Always **report the complete results of the first-stage regression**.
  - This includes all instrumental variables (IVs) and exogenous regressors used in the equation.
- Coefficients should have the expected sign and magnitude.
- Report the standard errors for these estimates to ensure statistical significance (Ebbes, Papies, and Van Heerde 2011).

Evaluating Instrument Strength

- **Report the R^2** of the first-stage regression, both with and without the IVs included.
  - A large difference in R^2 indicates strong instruments[1].
- Report the **F-statistic** for the excluded IVs.
  - A common benchmark is $F > 10$ for strong instruments, but this should be interpreted as a minimum requirement (Stock and Yogo 2002; Rossi 2014b).

1. **Weak Instrument:** An instrument that has a weak correlation with the endogenous variable, leading to unreliable estimates in instrumental variable (IV) regressions.

Practical Guidelines for Using TSLS

Stability and Sensitivity Analysis

- If multiple IVs are used for a single endogenous regressor, compare results:
  - Use the theoretically strongest IV alone and compare it to using all IVs.
  - This helps check the stability of the results. If large differences are found, the validity of the IVs should be re-examined (Angrist and Pischke 2008; Ebbes, Papies, and Van Heerde 2011).

Endogeneity Testing

- Perform a formal test for the presence of endogeneity, such as the **Hausman test**.
  - This test compares OLS estimates to 2SLS and checks if the OLS estimator is consistent.
  - If the null hypothesis is rejected, then 2SLS should be used (Wooldridge 2010; Verbeek 2012).

IV Approach - Hands on

▶ Code

IV-2SLS Estimation Summary			
Dep. Variable:	Y	R-squared:	0.9316
Estimator:	IV-2SLS	Adj. R-squared:	0.9314
No. Observations:	1000	F-statistic:	4268.1
Date:	Mon, Sep 16 2024	P-value (F-stat)	0.0000
Time:	12:50:10	Distribution:	F(2,997)
Cov. Estimator:	homoskedastic		

▶ Code

Parameter Estimates					
	Parameter	Std. Err.	T-stat	P-value	Lower CI
const	2.0060	0.0323	62.064	0.0000	1.9426
X	3.0222	0.0330	91.695	0.0000	2.9575
W	1.4603	0.1262	11.574	0.0000	1.2128

IV Approach - Hands on

Wu-Hausman-Test

▶ Code

```
Wu-Hausman test of exogeneity
H0: All endogenous variables are exogenous
Statistic: 123.2194
P-value: 0.0000
Distributed: F(1,996)
WaldTestStatistic, id: 0x255a35ef1d0
```

Wooldridge-Test

▶ Code

```
Wooldridge's regression test of exogeneity
H0: Endogenous variables are exogenous
Statistic: 123.2194
P-value: 0.0000
Distributed: chi2(1)
WaldTestStatistic, id: 0x255a35f4990
```

$\Rightarrow$ **Sargan-Test** not possible in our 1-Instrument setting!

Control Function Approach

- **Purpose**: A method to correct for endogeneity by incorporating an additional "control function" in the model to account for endogenous explanatory variables ((Papies, Ebbes, and Van Heerde 2017), (Ebbes, Papies, and Van Heerde 2011) and (Petrin and Train 2010)).
  - For the linear model (OLS), the CF approach is **exactly equivalent to TSLS**!

💡 Idea: Control Function Approach (CF)

- A control function is constructed using the residuals from the endogenous variable's first-stage regression (Petrin and Train 2010).
- This function is then included in the second-stage regression to purge the correlation between the endogenous regressor and the error term (Hill et al. 2021).

Control Function Approach

Key Concepts:

1 First Stage:

- Regress the endogenous variable on all exogenous variables (including instruments).
- Obtain the residuals (control function) (Petrin and Train 2010).

2 Second Stage:

- Include the control function from the first stage in the original model (Wooldridge 2015).
- Estimate the regression with the control function to account for endogeneity (Angrist and Krueger 2001).

Control Function Approach

Lets look at a simple OLS Example with one endogenous variable:

$$Y = \alpha + \beta X + \gamma W + \xi$$

Where

- X is exogenous (holding orthogonality condition $\mathbb{E}[X'\xi] = 0$) and can be used as valid instrument.
- W is an endogenous variable.
- and ξ as the errorterm.

Now, we will follow the **Stages** for the Control Function approach.

Control Function Approach

1 First Stage

Regress W on all exogenous variables X and instruments Z:

$$W = \underbrace{\pi_0 + \pi_1 Z}_{\substack{\text{Part of } W \text{ that is} \\ \text{not correlated with } \xi}} + \underbrace{\zeta}_{\substack{\text{Part of } W \text{ that is} \\ \text{correlated with } \xi}}$$

Where:

- $\mathbb{E}[Z'\zeta] = 0$.
- W is the endogenous variable.
- Z is the instruments
- ζ are the residuals (**CF**).

We can write the linear projection of ξ on ζ in error form as

$$\xi = \lambda\zeta + \varepsilon$$

Control Function Approach

2 Second Stage:

Plugging the linear projection into the model:

$$Y = \alpha + \beta X + \lambda\zeta + \varepsilon$$

Where we see ζ as an (exogenous) explanatory variable and $\lambda\zeta$ is the control function term to account for endogeneity (Papies, Ebbes, and Van Heerde 2017).

...

$\Rightarrow$ Kim and Petrin (2022) show that projection and rewriting would lead to the same (numerical!) form of IV approach (**only for linear models with additive errors**).

Control Function Approach - Hands on

▶ Code

OLS Regression Results			
Dep. Variable:	Y	R-squared:	0.980
Model:	OLS	Adj. R-squared:	0.980
Method:	Least Squares	F-statistic:	1.598e+04
Date:	Mon, 16 Sep 2024	Prob (F-statistic):	0.00
Time:	12:50:11	Log-Likelihood:	-830.78
No. Observations:	1000	AIC:	1670.
Df Residuals:	996	BIC:	1689.
Df Model:	3		
Covariance Type:	nonrobust		

▶ Code

	coef	std err	t	P>|t|	[0.025
Intercept	2.0060	0.018	113.773	0.000	1.971
X	3.0222	0.018	168.092	0.000	2.987
W	1.4603	0.069	21.217	0.000	1.325
residuals_W	0.7848	0.071	11.100	0.000	0.646

Copulas

- We've looked at IV Methods so far, but what if we can't find a valid instrument?
  - IV-free Methods: **Copulas**
- **Copulas** decompose a joint probability distribution (Note 3) into:
  - **Marginals**: individual distributions, each with no inherent correlation, see Note 2.
  - **Copula**: a coupling function that links the marginals, capturing the dependency structure.

> **Important**
>
> **The copula allows us to specify the correlation between variables separately from their marginal distributions!**

Copulas and Endogeneity

- One of the primary applications of (gaussian) copulas is in addressing endogeneity, particularly in econometrics and marketing models.
- (Gaussian) copula can be used to model the correlation between endogenous variables and the error term.
  - This is achieved by linking the non-normally distributed endogenous variable to a normally distributed error term through a copula structure.
- The approach from Park and Gupta (2012) introduces a Gaussian copula correction term to the regression model.
  - Suitable for management research where valid IVs are scarce.

The Gaussian Copula Approach (GC)

- The model is the same es before:

$$Y = \alpha + \beta X + \gamma W + \xi$$

- **Sklar's Theorem**:
  - Any joint distribution (Note 3) can be decomposed into marginal distributions (Note 2) and a copula, which captures the dependency between them.

> **Important**
>
> Let W be an endogenous regressor and ξ be a structural error term. According to **Sklar's Theorem** (Sklar, 1959), there exists a copula function $C(\cdot, \cdot)$ such that the joint cumulative distribution function (Note 1) of W and ξ can be written as:
>
> $$F(W, \xi) = C(H(W), G(\xi)) = C(U_W, U_\xi)$$
>
> where $U_W = H(W)$ and $U_\xi = G(\xi)$ are the marginal CDFs (Note 2) of W and ξ, respectively, both following uniform distributions on $(0, 1)$.
>
> Thus, the copula $C(\cdot, \cdot)$ captures the dependence structure between W and ξ, allowing the marginal distributions (Note 2) of W and ξ to be modeled separately from their joint dependence.

The Gaussian Copula Approach (GC)

Park and Gupta (2012) propose a **generated regressor approach** for handling endogeneity using a copula-based transformation. Here's a concise description:

They suggest that to correct for the endogeneity of W, one can use the following method:

- 1. **Transformation via the Copula:** First, transform the endogenous variable W into a uniform random variable U_W using its marginal CDF (Note 2), i.e., $U_W = H(W)$, where $H(\cdot)$ is the CDF of W.
- 2. **Apply the Inverse Normal CDF:** Then, generate a new variable $W^* = \Phi^{-1}(U_W)$, where $\Phi^{-1}(\cdot)$ is the inverse of the standard normal cumulative distribution function (Note 1). This transformation results in a normally distributed variable.
- 3. **OLS Estimation:** Finally, include W^* as an additional regressor in the OLS estimation of the structural equation. This step helps correct for the endogeneity of W, as including the transformed regressor W^* allows for consistent estimation of the model parameters.

The Gaussian Copula Approach (GC)

- **Identifying Assumption**:
  - The potentially endogenous regressor follows a **non-normal distribution**.
  - The error term follows a **normal distribution**.
- **GC Estimation Process** [1]:
  - Adds an additional term to the regression equation:

$$Y = \alpha + \beta X + \gamma W + \gamma^* W^* + \xi^*$$

  - $W^* = \Phi^{-1}(U_W)$, where Φ^{-1} is the inverse normal CDF (Note 1) and U_W is the empirical CDF of W.

1. Park and Gupta (2012)

The Gaussian Copula Approach - Hands on

- To illustrate the approach, lets construct data with non-normal distributed endogenous regressor:

$$
\begin{aligned}
X_{\text{exog}} &\sim \Gamma(1, 0.5) \\
\epsilon_1 &\sim \mathcal{N}(0, 0.2^2) \\
\epsilon_2 &\sim \mathcal{N}(0, 0.8^2)
\end{aligned}
$$

With an endogenous variable X and outcome variable Y as:

$$
\begin{aligned}
X &= X_{\text{exog}} + \epsilon_1 \\
Y &= \alpha + \beta X + (\epsilon_1 + \epsilon_2)
\end{aligned}
$$

with parameters $\alpha = 2$, $\beta = 1.5$. Here, ϵ_1 induces endogeneity in X and Y, while ϵ_2 introduces additional noise to the outcome.

This structure demonstrates the influence of both the exogenous component and the shared error term in X and Y.

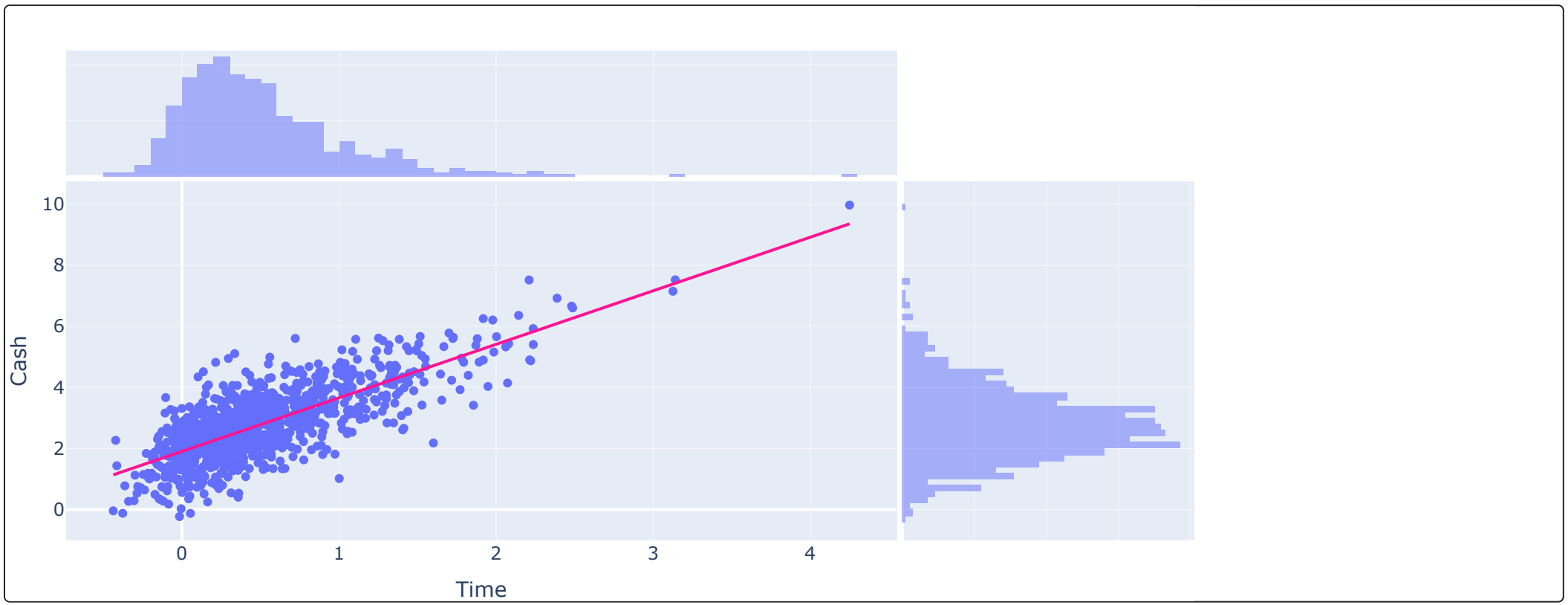
The Gaussian Copula Approach - Hands on

▶ Code



The Gaussian Copula Approach - Hands on

▶ Code

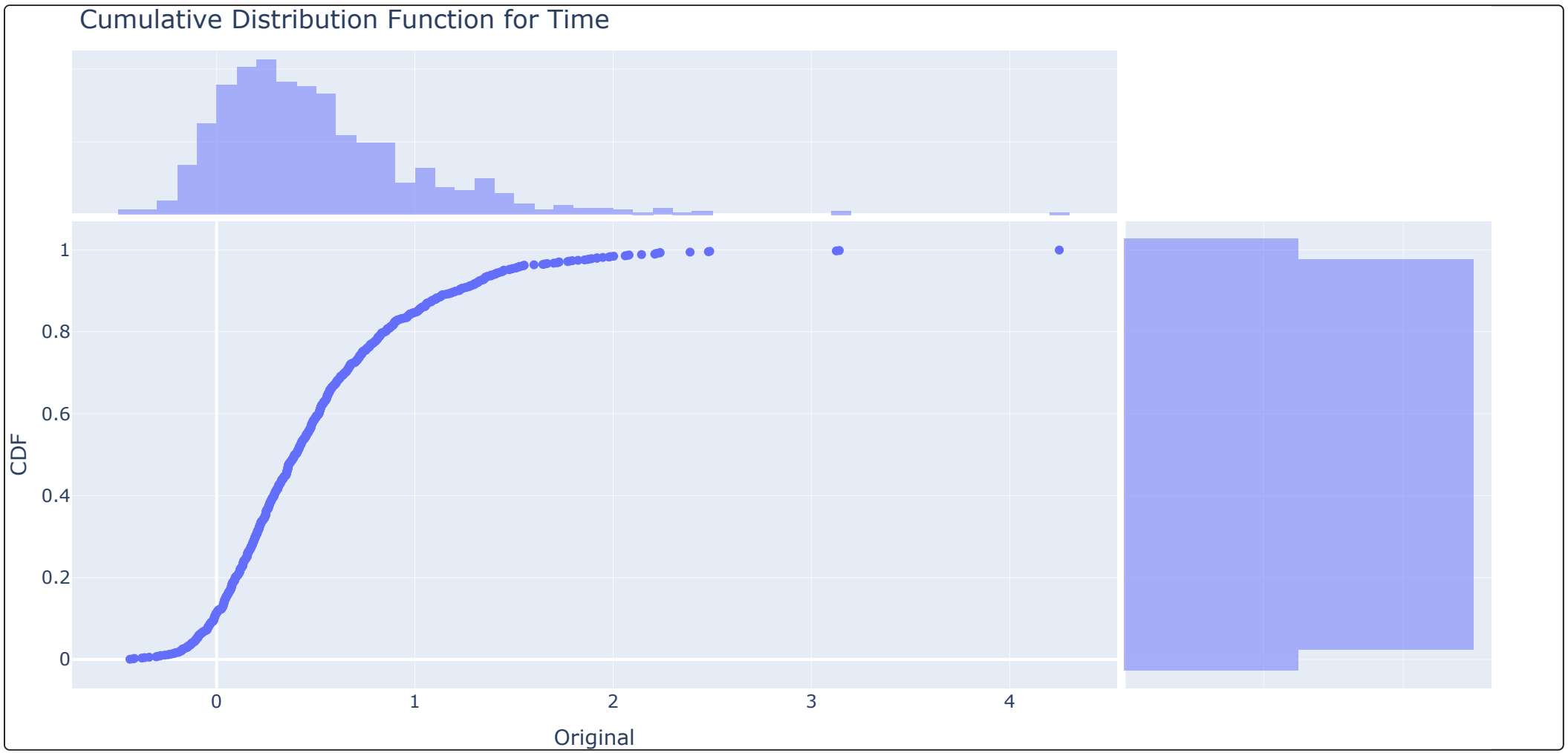


▶ Code

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	1.9019	0.035	54.594	0.000	1.834	1.970
Time	1.7547	0.048	36.419	0.000	1.660	1.849

The Gaussian Copula Approach - Hands on

▶ Code



The Gaussian Copula Approach - Hands on

▶ Code

OLS Regression Results			
Dep. Variable:	Cash	R-squared:	0.572
Model:	OLS	Adj. R-squared:	0.571
Method:	Least Squares	F-statistic:	665.9
Date:	Mon, 16 Sep 2024	Prob (F-statistic):	2.12e-184
Time:	12:50:11	Log-Likelihood:	-1176.8
No. Observations:	1000	AIC:	2360.
Df Residuals:	997	BIC:	2374.
Df Model:	2		
Covariance Type:	nonrobust		

▶ Code

	coef	std err	t	P>|t|	[0.025	0.
Intercept	2.0253	0.080	25.319	0.000	1.868	2.
Time	1.5108	0.150	10.055	0.000	1.216	1.
Time_gc	0.1331	0.078	1.713	0.087	-0.019	0.

Thank you for your attention  !

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Appendix

Appendix - Measurement Error

Measurement Error

The relationship between education and wages can be modeled (e.g. simplified *Mincer Equation*) as:

$$\ln(W) = \alpha + \beta E + u$$

Where:

- W is the wage,
- E is the years of education,
- u is the error term,
- β is the return to education (i.e., the effect of an additional year of education on wages).

Measurement Error

If education E is measured with error, say we observe E^* instead of the true value E, then:

$$E^* = E + v$$

Where:

- E^* is the observed (but erroneous) measure of education,
- v is the measurement error.

This leads to the observed equation:

$$\ln(W) = \alpha + \beta(E^* - v) + u$$

Measurement Error

Rearranging the previous equation gives:

$$\ln(W) = \alpha + \beta E^* + (\beta v + u)$$

Here, the new error term $(\beta v + u)$ consists of both the original error u and the measurement error v. This causes **endogeneity** because the measurement error v is likely correlated with the observed E_i^*, violating the assumption that the error term is uncorrelated with the independent variable, leading to biased and inconsistent estimates of β.

Appendix: A short simulation 

Simulation Data

We define the following DGP:

$$
\begin{aligned}
X &\sim \mathcal{N}(0, 1) \\
Z &\sim \mathcal{N}(0, 1) \\
\xi &\sim \mathcal{N}(0, 1)
\end{aligned}
$$

$$
W = \mathrm{snrW} \times (\beta_Z Z + \beta_\xi \xi) + (1 - \beta_Z - \beta_\xi)\epsilon, \quad \epsilon \sim \mathcal{N}(0, 1)
$$

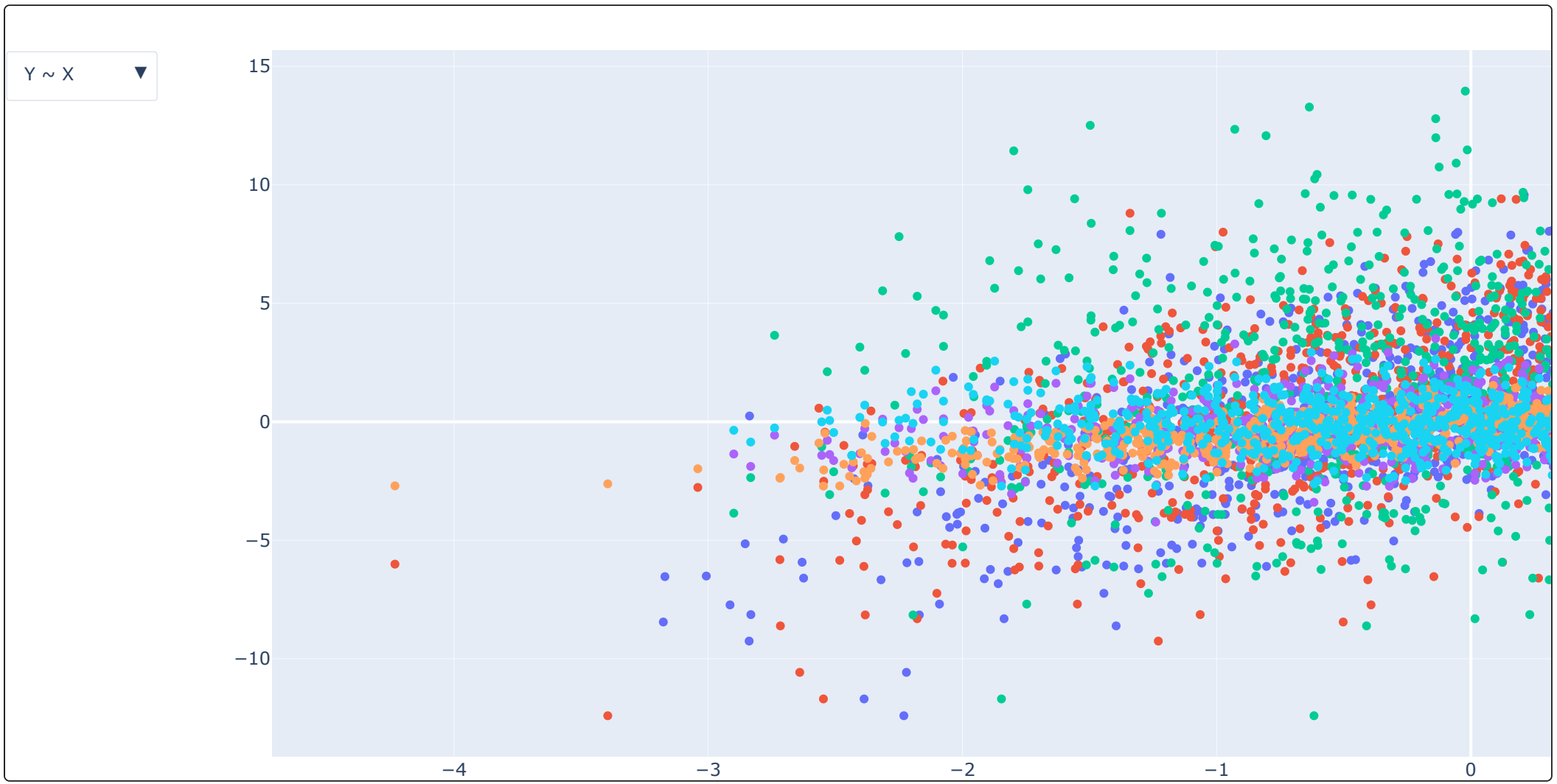
$$
Y = \alpha + \beta X + \gamma W + \xi
$$

with $\alpha = 2, \beta = 3, \gamma = 1.5, \beta_Z = 0.1, \beta_\xi = 0.3$, and $\mathrm{snrW} = 3$.

▶ Code

Simulation Data

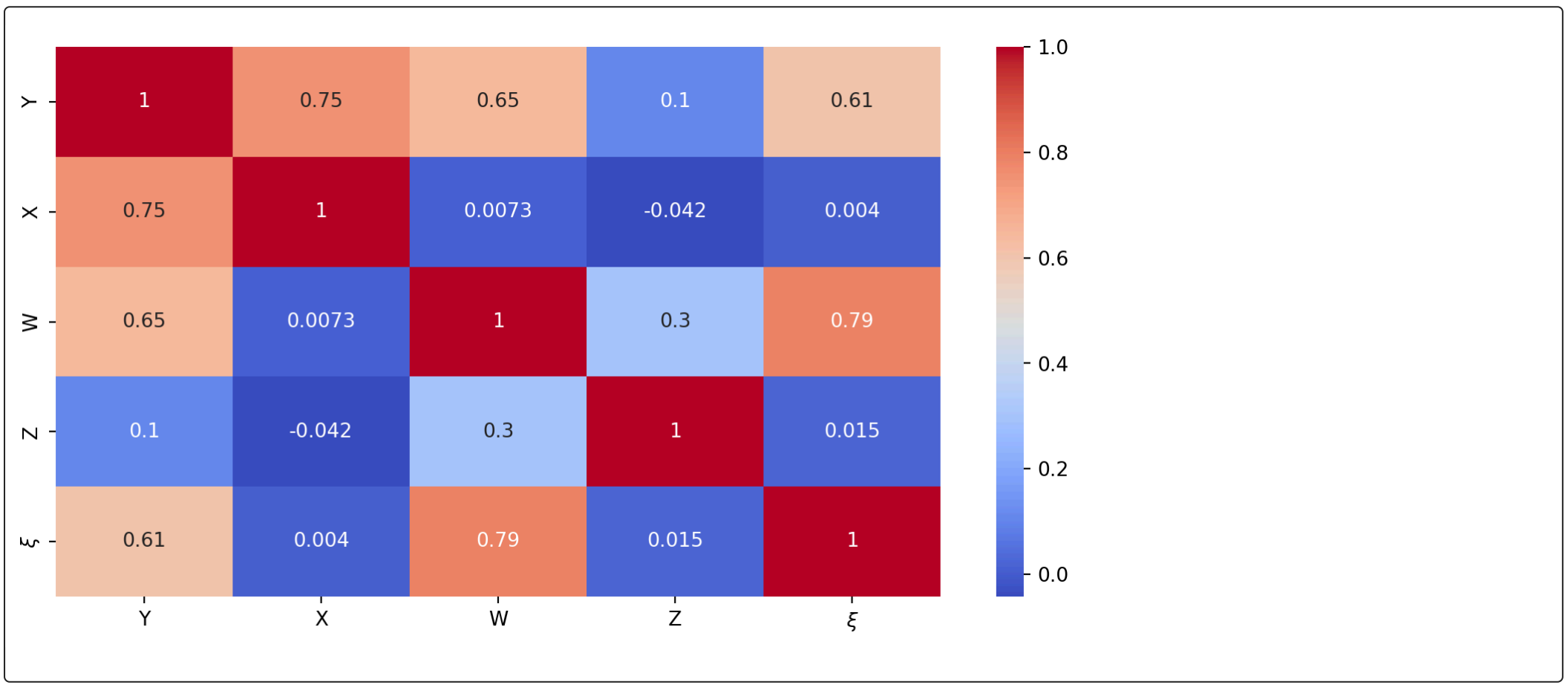
▶ Code



Simulation Data

We can plot a correlation matrix as a heatmap:

▶ Code



OLS Approach

▶ Code

OLS Regression Results			
Dep. Variable:	Y	R-squared:	0.976
Model:	OLS	Adj. R-squared:	0.976
Method:	Least Squares	F-statistic:	1.991e+04
Date:	Mon, 16 Sep 2024	Prob (F-statistic):	0.00
Time:	12:50:12	Log-Likelihood:	-946.24
No. Observations:	1000	AIC:	1898.
Df Residuals:	997	BIC:	1913.
Df Model:	2		
Covariance Type:	nonrobust		

▶ Code

	coef	std err	t	P>|t|	[0.025	0
Intercept	1.9912	0.020	100.714	0.000	1.952	2
X	2.9982	0.020	150.642	0.000	2.959	3
W	2.1976	0.017	129.742	0.000	2.164	2

IV Approach

▶ Code

OLS Regression Results			
Dep. Variable:	Y	R-squared:	0.582
Model:	OLS	Adj. R-squared:	0.581
Method:	Least Squares	F-statistic:	693.0
Date:	Mon, 16 Sep 2024	Prob (F-statistic):	2.26e-189
Time:	12:50:12	Log-Likelihood:	-2366.5
No. Observations:	1000	AIC:	4739.
Df Residuals:	997	BIC:	4754.
Df Model:	2		
Covariance Type:	nonrobust		

▶ Code

	coef	std err	t	P>|t|	[0.025	0
Intercept	1.9732	0.082	24.052	0.000	1.812	2.
X	3.0038	0.082	36.461	0.000	2.842	3.
W_hat	1.5437	0.232	6.645	0.000	1.088	2.

Reporting first stage results:

▶ Code

```
R² with IV: 0.09109572888942097
R² without IV: 5.268057206464505e-05
F-statistic for the excluded instrument (Z): 99.86741404734919
```

IV Approach - Using pythons `linearmodels` API

▶ Code

IV-2SLS Estimation Summary			
Dep. Variable:	Y	R-squared:	0.9390
Estimator:	IV-2SLS	Adj. R-squared:	0.9389
No. Observations:	1000	F-statistic:	4756.9
Date:	Mon, Sep 16 2024	P-value (F-stat)	0.0000
Time:	12:50:12	Distribution:	F(2,997)
Cov. Estimator:	homoskedastic		

▶ Code

Parameter Estimates					
	Parameter	Std. Err.	T-stat	P-value	Lower CI
const	1.9732	0.0313	63.016	0.0000	1.9118
X	3.0038	0.0314	95.526	0.0000	2.9421
W	1.5437	0.0887	17.410	0.0000	1.3697

IV Approach - Using pythons `linearmodels` API: Testing

Wu-Hausman-Test

▶ Code

```
Wu-Hausman test of exogeneity
H0: All endogenous variables are exogenous
Statistic: 175.3970
P-value: 0.0000
Distributed: F(1,996)
WaldTestStatistic, id: 0x255a34ef010
```

Wooldridge-Test

▶ Code

```
Wooldridge's regression test of exogeneity
H0: Endogenous variables are exogenous
Statistic: 175.3970
P-value: 0.0000
Distributed: chi2(1)
WaldTestStatistic, id: 0x255a34ee890
```

$\Rightarrow$ **Sargan-Test** not possible in our 1-Instrument setting!

IV - Control Function Approach

▶ Code

OLS Regression Results			
Dep. Variable:	Y	R-squared:	0.979
Model:	OLS	Adj. R-squared:	0.979
Method:	Least Squares	F-statistic:	1.565e+04
Date:	Mon, 16 Sep 2024	Prob (F-statistic):	0.00
Time:	12:50:12	Log-Likelihood:	-865.14
No. Observations:	1000	AIC:	1738.
Df Residuals:	996	BIC:	1758.
Df Model:	3		
Covariance Type:	nonrobust		

▶ Code

	coef	std err	t	P>|t|	[0.025
Intercept	1.9732	0.018	107.888	0.000	1.937
X	3.0038	0.018	163.547	0.000	2.968
W	1.5437	0.052	29.808	0.000	1.442
residuals_W	0.7194	0.054	13.244	0.000	0.613

Appendix: But what about non-linear models  ?

IV in non-linear Models 🧙

- Assumptions on linearity may not hold for all parts of a SEM!
  - One can assumpe for example partially-linear relationships.
- OLS and linear models (parametric models in general) have a number of comparatively strict assumptions.
  - If these are not fulfilled, you should use semi-parametric or non-parametric models.

$\Rightarrow$ **Lets create data with non-linear respones and run the simulation again!**

Non-linear Simulation Data

We define the following non-linear DGP:

$$
\begin{aligned}
X &\sim \mathcal{N}(0, 1) \\
Z &= 0.8 \cdot X^2 + 0.2 \cdot \mathcal{N}(0, 1) \\
W &= \beta_Z \cdot Z + \beta_X \cdot \cos(X^2) + \beta_\xi \cdot \xi + 0.5 \cdot \mathcal{N}(0, 1) \\
\xi &\sim \mathcal{N}(0, 1)
\end{aligned}
$$

Here, W is correlated with the error term ξ, hence endogenous. The outcome variable Y is defined as:

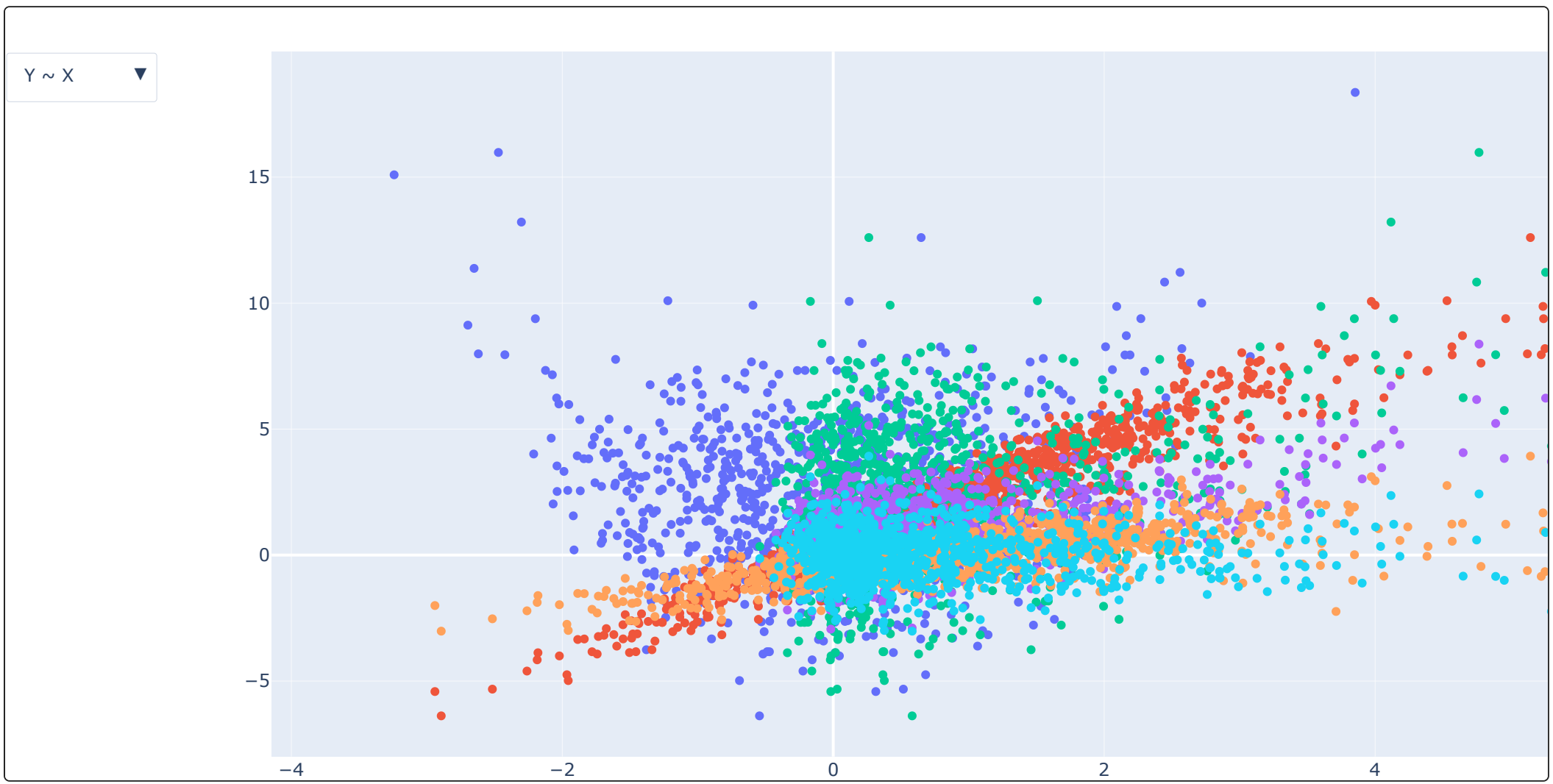
$$
Y = \alpha + \beta \cdot \cos(X^2) + \gamma \cdot W + \xi
$$

with $\alpha = 0.5, \beta = 0.5, \gamma = 1.5, \beta_Z = 1, \beta_\xi = 1$.

▶ Code

Non-linear Simulation Data

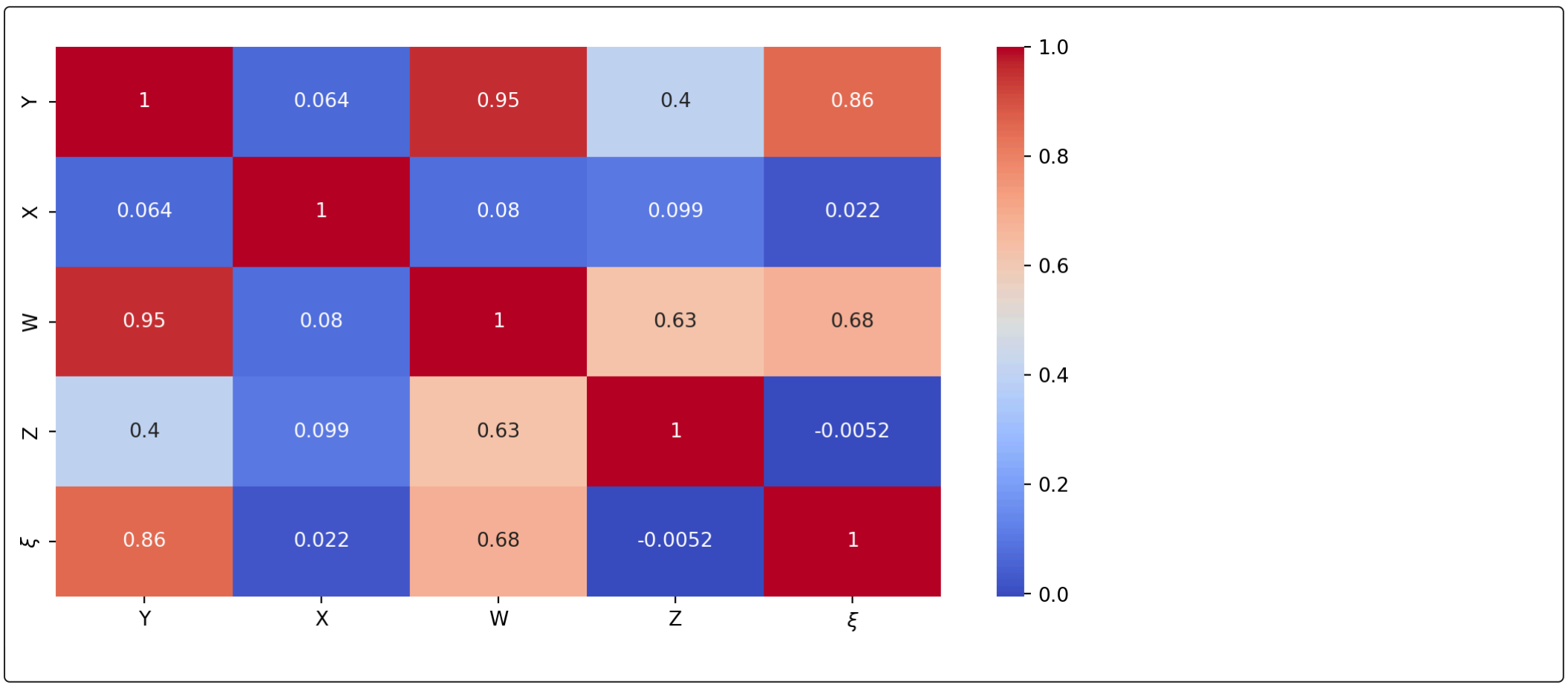
▶ Code



Non-linear Simulation Data

We can plot a correlation matrix as a heatmap:

▶ Code



OLS Approach for non-linearity

▶ Code

OLS Regression Results			
Dep. Variable:	Y	R-squared:	0.911
Model:	OLS	Adj. R-squared:	0.910
Method:	Least Squares	F-statistic:	5073.
Date:	Mon, 16 Sep 2024	Prob (F-statistic):	0.00
Time:	12:50:13	Log-Likelihood:	-1270.7
No. Observations:	1000	AIC:	2547.
Df Residuals:	997	BIC:	2562.
Df Model:	2		
Covariance Type:	nonrobust		

▶ Code

	coef	std err	t	P>|t|	[0.025
Intercept	0.3812	0.034	11.257	0.000	0.315
X	-0.0346	0.028	-1.237	0.216	-0.090
W	1.8892	0.019	100.500	0.000	1.852

IV Approach for non-linearity

▶ Code

OLS Regression Results			
Dep. Variable:	Y	R-squared:	0.160
Model:	OLS	Adj. R-squared:	0.159
Method:	Least Squares	F-statistic:	95.19
Date:	Mon, 16 Sep 2024	Prob (F-statistic):	1.46e-38
Time:	12:50:13	Log-Likelihood:	-2390.2
No. Observations:	1000	AIC:	4786.
Df Residuals:	997	BIC:	4801.
Df Model:	2		
Covariance Type:	nonrobust		

▶ Code

	coef	std err	t	P>|t|	[0.025	0.
Intercept	1.0540	0.129	8.163	0.000	0.801	1.
X	0.0404	0.086	0.469	0.639	-0.129	0.
W_hat	1.2574	0.092	13.618	0.000	1.076	1.

Reporting first stage results:

▶ Code

```
R² with IV: 0.3928631561149165
R² without IV: 0.006362466483357809
F-statistic for the excluded instrument (Z): 634.6858890935622
```

IV for non-linearity - Using pythons `linearmodels` API

▶ Code

IV-2SLS Estimation Summary			
Dep. Variable:	Y	R-squared:	0.8092
Estimator:	IV-2SLS	Adj. R-squared:	0.8088
No. Observations:	1000	F-statistic:	418.87
Date:	Mon, Sep 16 2024	P-value (F-stat)	0.0000
Time:	12:50:14	Distribution:	F(2,997)
Cov. Estimator:	homoskedastic		

▶ Code

Parameter Estimates					
	Parameter	Std. Err.	T-stat	P-value	Lower CI
const	1.0540	0.0615	17.124	0.0000	0.9332
X	0.0404	0.0411	0.9837	0.3255	-0.0402
W	1.2574	0.0440	28.567	0.0000	1.1710

IV Approach for non-linearity - Testing

Wu-Hausman-Test

▶ Code

```
Wu-Hausman test of exogeneity
H0: All endogenous variables are exogenous
Statistic: 2576.1660
P-value: 0.0000
Distributed: F(1,996)
WaldTestStatistic, id: 0x255a3954590
```

Wooldridge-Test

▶ Code

```
Wooldridge's regression test of exogeneity
H0: Endogenous variables are exogenous
Statistic: 2576.1660
P-value: 0.0000
Distributed: chi2(1)
WaldTestStatistic, id: 0x255a3a1aa50
```

$\Rightarrow$ **Sargan-Test** not possible in our 1-Instrument setting!

IV for non-linearity - Control Function Approach

▶ Code

OLS Regression Results			
Dep. Variable:	Y	R-squared:	0.975
Model:	OLS	Adj. R-squared:	0.975
Method:	Least Squares	F-statistic:	1.298e+04
Date:	Mon, 16 Sep 2024	Prob (F-statistic):	0.00
Time:	12:50:14	Log-Likelihood:	-632.08
No. Observations:	1000	AIC:	1272.
Df Residuals:	996	BIC:	1292.
Df Model:	3		
Covariance Type:	nonrobust		

▶ Code

	coef	std err	t	P>|t|	[0.025
Intercept	1.0540	0.022	47.336	0.000	1.010
X	0.0404	0.015	2.719	0.007	0.011
W	1.2574	0.016	78.969	0.000	1.226
residuals_W	1.0339	0.020	50.756	0.000	0.994

IV for non-linear Models 🧙

💡 **Idea: Partially Linear IV (PLIV) model**

- Use of flexible machine learning methods (e.g., random forests, neural networks) in the first stage to capture potentially complex, non-linear relationships between endogenous variables and instruments. This makes it more robust in cases where the relationships are not strictly linear.
- Controls for **1. Overfitting** and ensuring **2. Valid Inference**.
  - Achieves this by **1. Cross-Fitting** and by solving **2. Neyman Orthogonal Scores**.
  - **TSLS** is straightforward and **based on traditional asymptotic theory**, but it can be unreliable in complex or high-dimensional settings.

IV for non-linear Models

- The **Partially Linear IV (PLIV) model** takes the following form:

$$Y = D\theta_0 + g_0(X) + \xi, \qquad E[\xi \mid Z, X] = 0$$
$$Z = m_0(X) + V, \qquad E[V \mid X] = 0$$

Where:

- Y is the outcome variable,
- D is the variable of interest (endogenous),
- Z denotes one or multiple instrumental variables,
- $X = (X_1, \ldots, X_p)$ is a exogenous vector consisting of other covariates,
- ξ and V are stochastic errors.

- The the second stage solves the **Neyman Orthogonal Score**:

$$\begin{aligned}\psi(W; \theta, \eta) &:= [Y - \ell(X) - \theta(D - r(X))][Z - m(X)] \\ &= -(D - r(X))(Z - m(X))\theta + (Y - \ell(X))(Z - m(X)) \\ &= \psi_a(W; \eta)\theta + \psi_b(W; \eta)\end{aligned}$$

IV in non-linear Models

Lets use the `DoubleML` package to solve our Simulation Example:

▶ Code

	coef	std err	t	P>|t|	2.5 %	97.5 %
W	1.55831	0.049646	31.388346	2.918284e-216	1.461005	1.655614

We can access the R^2 Scores of the learners:

▶ Code

```
{'ml_l': array([[0.04204007]]),
 'ml_m': array([[0.75985982]]),
 'ml_r': array([[0.19149137]])}
```

Appendix: Double Machine Learning

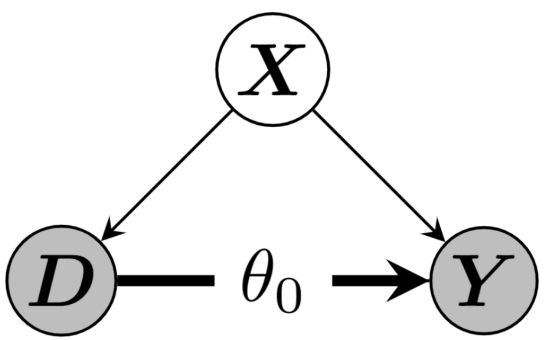
Motivating Example - PLR

- Revisit the PLR model from the beginning:

$$
\begin{aligned}
Y &= \theta_0 D + g_0(X) + \varepsilon, & \mathbb{E}[\varepsilon | X, D] &= 0 \\
D &= m_0(X) + \vartheta, & \mathbb{E}[\vartheta | X] &= 0
\end{aligned}
$$

with

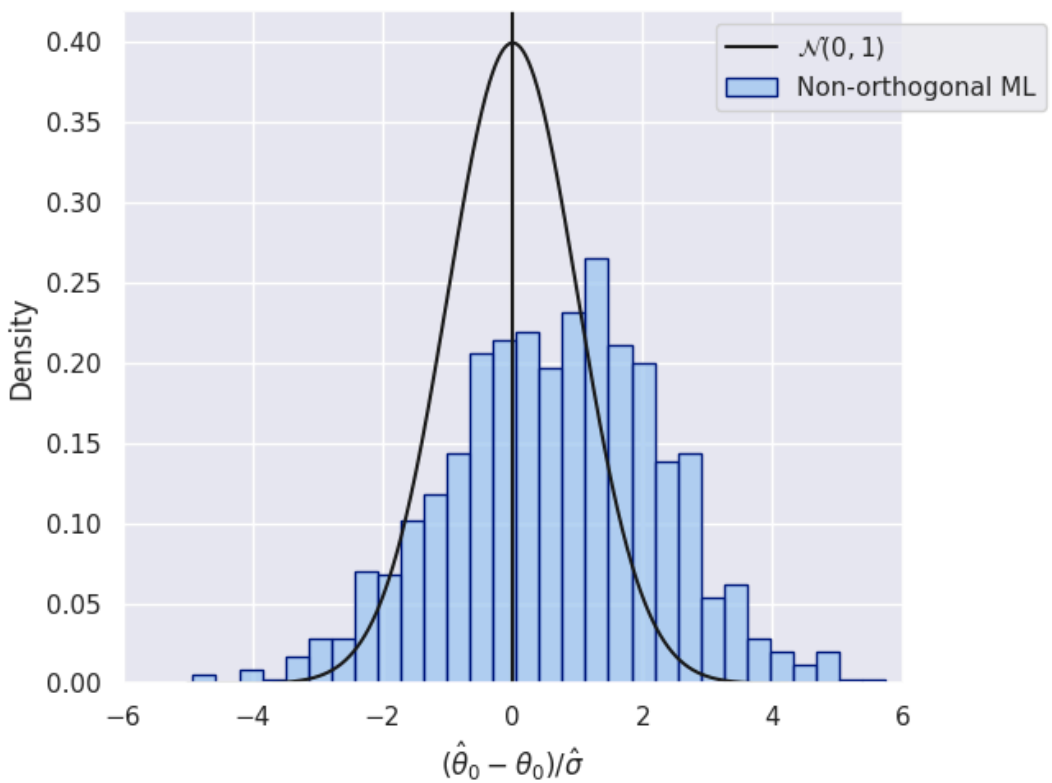
- Y - outcome variable
- D - policy/treatment variable
- $X = (X_1, \ldots X_p)^T$ - vector of additional controls
- ε, ϑ - stochastic errors



DAG for PLR Model

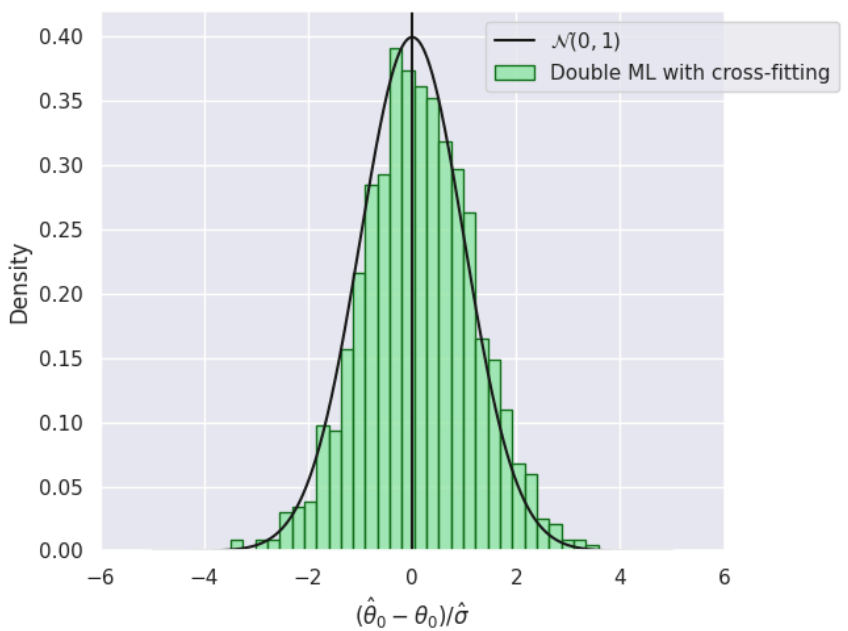
Motivating Example - Regularization Bias

- What if we simply plug-in ML predictions $\hat{g}(X)$ for $g_0(X)$ into $Y = \theta_0 D + g_0(X) + \varepsilon$?



Motivating Example - Orthogonalization

- *Frisch-Waugh-Lovell* style approach: θ_0 can be consistently estimated by **partialling out** X, i.e,
  1. Predict Y and D by $\mathbb{E}[Y|X]$ and $\mathbb{E}[D|X]$, obtained using ML methods
  2. Residualize $\tilde{Y} = Y - \mathbb{E}[Y|X]$ and $\tilde{D} = D - \mathbb{E}[D|X]$
  3. Regress $\tilde{Y}$ on $\tilde{D}$ to obtain $\hat{\theta}$



Double/Debiased Machine Learning (DML)

- **Neyman Orthogonality**

$$\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta)]|_{\eta=\eta_0} = 0$$

  ensures that the moment condition identifying θ_0 is **insensitive to small pertubations** of the nuisance function η around η_0

- Assumptions for the nuisance elements (see Chernozhukov et al. (2018)) *[Math Processing Error]*
- Under some regularity conditions the estimator $\hat{\theta}$ concentrates in $1/\sqrt{N}$-neighborhood of θ_0 and

$$\sqrt{N}(\hat{\theta} - \theta_0) \to \mathcal{N}(0, \sigma^2)$$

$\Rightarrow$ simple approach to control for confounders, *if they can be used for ML models!*

Neyman Orthogonality

The naive approach minimizes the following MSE

$$min_\theta \mathbb{E}[(Y - D\theta_0 - g_0(X))^2]$$

This implies the following moment equation

$$\mathbb{E}[\underbrace{(Y - D\theta_0 - g_0(X))D}_{=:\psi(W,\theta_0,\eta)}] = 0$$

Whereas for the partialling-out approach minimizes

$$min_\theta \mathbb{E}\left[\left(Y - \mathbb{E}[Y|X] - (D - \mathbb{E}[D|X])\theta\right)^2\right]$$

which implies

$$\mathbb{E}\Big[\underbrace{\left(Y - \mathbb{E}[Y|X] - (D - \mathbb{E}[D|X])\theta\right)(D - \mathbb{E}[D|X])}_{=:\psi(W,\theta_0,\eta)}\Big] = 0$$

Neyman Orthogonality

Naive approach

$$\psi(W, \theta_0, \eta) = (Y - D\theta_0 - g_0(X))D$$

Regression adjustment score

$$\begin{aligned} \eta &= g(X), \\ \eta_0 &= g_0(X). \end{aligned}$$

FWL partialling out

$$\begin{aligned} \psi(W, \theta_0, \eta_0) = &\Big((Y - E[Y|X]) \\ &- (D - E[D|X])\theta_0\Big) \\ &(D - E[D|X]) \end{aligned}$$

Neyman-orthogonal score (Frisch-Waugh-Lovell)

$$\begin{aligned} \eta &= (\ell(X), m(X)), \\ \eta_0 &= (\ell_0(X), m_0(X)), \\ &= (\mathbb{E}[Y \mid X], \mathbb{E}[D \mid X]). \end{aligned}$$

DML Key Ingredients

1. Neyman Orthogonality

- Inference is based on a moment condition that satisfies the **Neyman orthogonality condition** $\psi(W; \theta, \eta)$

$$E[\psi(W; \theta_0, \eta_0)] = 0,$$

- where $W := (Y, D, X, Z)$ and with θ_0 being the unique solution that obeys the **Neyman orthogonality condition**

$$\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta)]|_{\eta=\eta_0} = 0.$$

- ∂_η denotes the pathwise (Gateaux) derivative operator

DML Key Ingredients

1. Neyman Orthogonality

- **Neyman orthogonality** ensures that the **moment condition** identifying θ_0 is **insensitive to small pertubations** of the nuisance function η around η_0
- Using a Neyman-orthogonal score **eliminates the first order biases** arising from the replacement of η_0 with a ML estimator $\hat{\eta}_0$
- PLR example: Partialling-out score function

$$\psi(\cdot) = (Y - E[Y|X] - \theta(D - E[D|X]))(D - E[D|X])$$

DML Key Ingredients

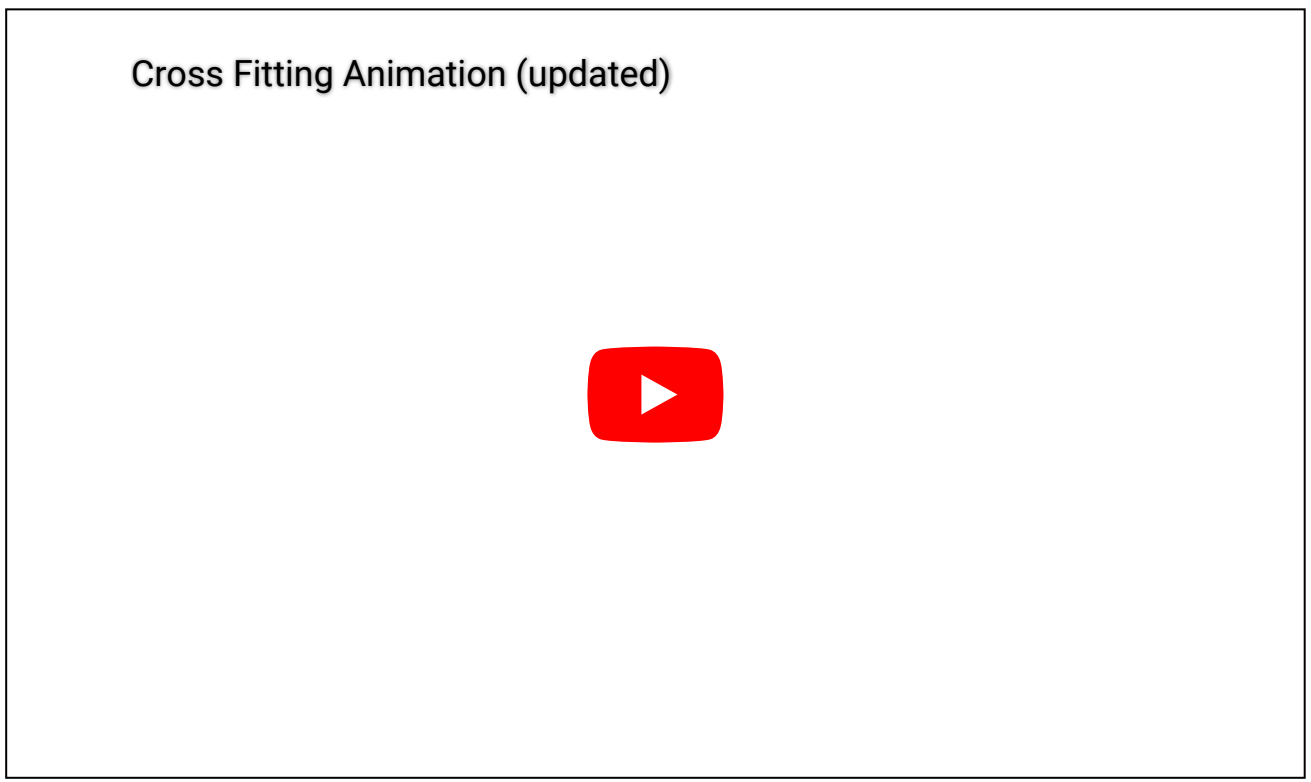
2. High-Quality Machine Learning Estimators

- The nuisance parameters are estimated with high-quality (fast-enough converging) machine learning methods.
- Different structural assumptions on η_0 lead to the use of different machine-learning tools for estimating η_0 Chernozhukov et al. (2018) (Section 3)
- Rate requirements depend on the causal model and orthogonal score, e.g. (see Chernozhukov et al. (2018)),
  - PLR, partialling out:
    $\|\hat{m}_0 - m_0\|_{P,2} \times \left(\|\hat{m}_0 - m_0\|_{P,2} + \|\hat{\ell}_0 - \ell_0\|_{P,2}\right) \leq \delta_N N^{-1/2}$
  - IRM/DR score, ATE: $\|\hat{m}_0 - m_0\|_{P,2} \times \|\hat{\ell}_0 - \ell_0\|_{P,2} \leq \delta_N N^{-1/2}$

DML Key Ingredients

3. Sample Splitting

- To avoid the biases arising from overfitting, a form of **sample splitting** is used at the stage of producing the estimator of the main parameter θ_0.
- Efficiency gains by using cross-fitting (swapping roles of samples for train / hold-out)



DML Key Ingredients

Main result in Chernozhukov et al. (2018)

> There exist regularity conditions, such that the DML estimator $\tilde{\theta}_0$ concentrates in a $1/\sqrt{N}$-neighborhood of θ_0 and the sampling error is approximately
>
> $$\sqrt{N}(\tilde{\theta}_0 - \theta_0) \sim N(0, \sigma^2),$$
>
> with
>
> $$\sigma^2 := J_0^{-2}\mathbb{E}(\psi^2(W; \theta_0, \eta_0)),$$
> $$J_0 = \mathbb{E}(\psi_a(W; \eta_0)).$$

- See this example based on Chernozhukov et al. (2018)

Appendix: Details to Distributions

Cumulative Distribution Function

> **Note**

- **Definition**: The CDF of a random variable X gives the probability that it takes on a value less than or equal to x. It provides the cumulative probability up to a specific point:

$$F(x) = P(X \leq x)$$

Marginal Distribution Function



Note

- **Definition**: The marginal distribution function of a random variable X in a multivariate setting is the CDF of X alone and describes its the distribution, ignoring the other variables in the joint distribution.:

$$F_X(x) = P(X \leq x)$$

Joint Distribution Function

> **Note**

- **Definition**: The joint distribution function of two or more random variables $X_1, X_2, \ldots, X_n$ gives the probability that each of the random variables takes on values less than or equal to specified points simultaneously:

$$F(x_1, x_2, \ldots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \ldots, X_n \leq x_n)$$